Week, 2nd Feb

Monday, 2 February 2026 10:32 AM

→ Simple model of elasticity

n_+ & n_- → orientation of monomer

$\Omega_{chain} = \frac{N!}{n_+! n_-!} = \frac{N!}{n_+!(N-n_+)!}$

→ Fundamental equation for elastic band

$dU = TdS + \tau\, dL + \mu dN$

tension ↓

$\tau \Leftrightarrow -p$

↑ acts opp. to pressure

$L \Leftrightarrow x$

→ As stretching out, the entropy goes down ↓

only limited arrangements for the microstates of monomers

$\left.\frac{\partial T}{\partial x}\right|_{S,N} > 0$

$\Omega(N,V,E) = \int_{E \le H \le E+\delta E} d\underline{p}\, d\underline{q}$ } is a volume in phase space in classical mechanics.

(Not dimensionless in this current eq.)

$\Delta p \Delta q \gtrsim \frac{h}{4\pi}$; $H = KE + pE$

p: position

Classical mechanics overcounts the microstates; but certain distances (less than h) cannot be measured.

for ideal gas: $\Omega(N,V,E) = V^N \int \cdots \int dp \cdots dp$

for ideal gas: $\Omega(N,V,E) = \frac{V^N}{h^{3N} N!} \int \cdots \int dp_1 \cdots dp_{3N}$

$0 \leq \frac{p^2}{2m} \leq E$

Gamma function,

Volume of n-dimensional hyperspace $\rightarrow V_N =$

Just from Hamiltonian & microstates $\rightarrow$ we derive everything

Canonical Ensemble (N, V, T) (Microstates vs H, V, E)

(Allows for energy exchange)

